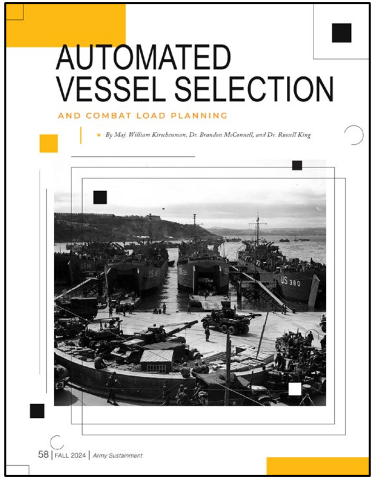
A Prioritized 2-D Packing Model for Military Combat Loading


AUTOMATED
VESSEL SELECTION
AND COMBAT LOAD PLANNING
By Maj. William Kirschenman, Dr. Brandon McConnell, and Dr. Russell King
58 | FALL 2024 | Army Sustainment

INFORMS Annual Meeting
Military and Security Track
October 28, 2025

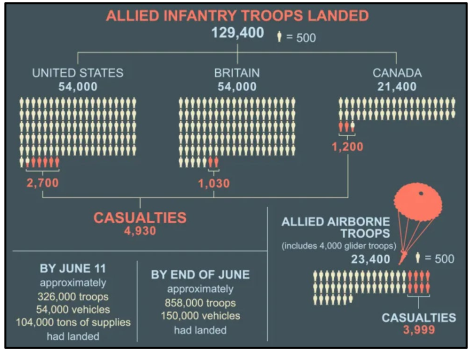
MAJ Will Kirschenman

Operations Research Ph.D. Candidate
North Carolina State University

Motivation

Normandy Invasion: June 6, 1944

- **Initial detailed planning** started in March 1943.
- **Combat loading** commenced in April 1944.
- Over 1,200 landing craft were loaded in 14 ports across southern England, each meticulously scheduled to avoid congestion on June 6, 1944.



Combat Loading Tenets

- **Tactical Sequencing.** Equipment must off-load in specific order for ground tactical plan.
- **Unit Integrity.** Military units fight as cohesive elements.
- **Asset Prioritization.** Critical equipment needs preferential placement.
- **Speed = Survivability.** Time spent off-loading is a high vulnerability period.

1ID Artillery Equipment Loading on LSTs at Brixham, England, on June 1, 1944.

Off-loading equipment on Omaha Beach in days following June 6, 1944.

Other Examples

- **Incheon Landing.**
- **Operation Cartwheel (Rabaul).**
- **Operation Husky (Invasion of Sicily).**
- **Operation Iceberg (Battle of Okinawa).**

What if we need to do this again?

Dissertation Overview

- One topic with three separate problem areas:
  - **Chapter 2:** A *generalized packing problem with prioritization*.
  - **Chapter 3:** Extending to a *single-vessel combat loading model*.
  - **Chapter 4:** Adding an *upstream vessel-selection problem for fleet optimization*.
- Successive chapters extend off each other.
- A proposed prioritization framework followed by military applications.

Central Research Question: *How can we automate the generation of combat load plans for a large landing force, explicitly incorporating multi-level prioritization of items and groups, critical operational constraints, and spatial layout objectives, thereby enhancing combat effectiveness?*

Paper 1: Prioritized 2-D Orthogonal Packing

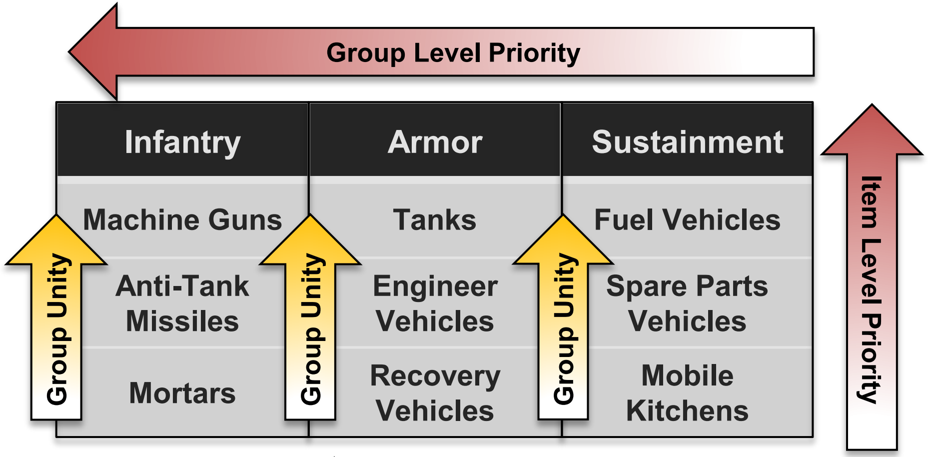
Formulation

Approach: Mixed-Integer Linear Program (MILP) based on efficient BPP formulations (Fontaine and Minner, 2023).

- Use a **Prioritization Matrix ($\pi_{i,k}$)** for spatial preferences:
  - **Diagonal ($\pi_{i,k}$ for $i = k$):** Priority weight for placing item *i* close to the bin access point (x^o, y^o). *Higher value = stronger pull to access point*.
  - **Off-Diagonal ($\pi_{i,k}$ for $i < k$):** Priority weight for placing items i and k (often same group) close to each other (cohesion). *Higher value = stronger pull together*.
- Optimize item placement by minimizing a **Weighted-Distance Objective Function** (rectilinear distance):

$$\min \sum_{i \in I} \sum_{k \in I, i \leq k} \pi_{ik} \left(\rho^x_{ik} + \rho^y_{ik} \right)$$

- **Key Constraint Categories:**
  - (1) Ensuring items fit within bin boundaries and reflect chosen orientation. (2) Preventing overlap between any pair of items. (3) Calculating rectilinear distance between item centroids and/or access point.



Key Takeaways

- Introduces a **novel 2-D orthogonal packing framework** that uniquely integrates multi-level spatial priorities.
- Demonstrates an **effective sliding-window matheuristic** that rapidly yields high-quality, priority-driven layouts, significantly outperforming direct MILP solutions.
- Established methods for **rigorous solution quality assessment** using new geometric lower bound and complementary statistical lower bound, enhancing confidence in our solution approaches.

Paper 1


The 2-D Orthogonal Packing Problem with Multiple Levels of Prioritization: A Spatial Optimization Perspective

Abstract

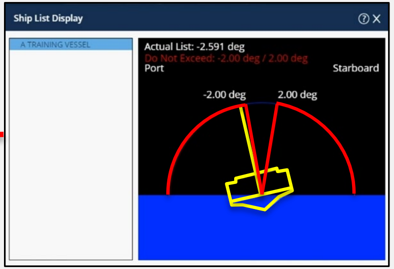
This paper addresses two-dimensional orthogonal packing within a confined space, integrating bin packing principles with facility layout concepts to address scenarios in which items must not only fit but also be arranged according to spatial priorities. We embed a prioritization matrix into the bin packing framework, enabling items to be clustered with one another or pulled toward certain bin access points based on assigned priority weights. Unlike traditional bin packing, which focuses on space utilization alone, our approach balances proximity to bin access points and adjacency among functionally related items, extending the utility of bin packing to applications requiring more nuanced layout preferences.

Add

Paper 2: Single Vessel Combat Loading

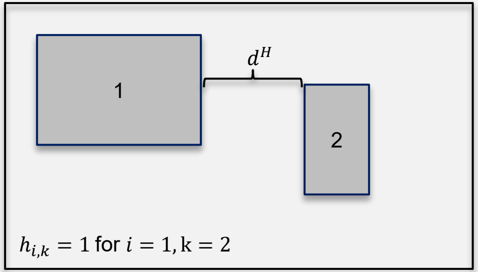
Operational Constraints Accounted for in Current Load Planning Systems (ICODES)

Load Balancing



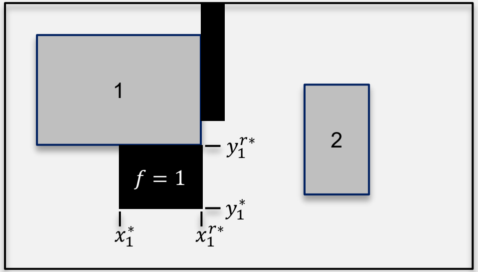
ICODES Single Load Planner: Ship list agent notifying the user of list (starboard-to-port) stability violations.

Material Separation



Example showing minimum material separation distance between a pair of items.

Obstacle Avoidance



Example showing obstacles (black fill) treated as fixed items and moveable items (gray fill) as "free" items.

Load Balancing: A Global Constraint

Challenge: *Load balancing is a global constraint* (depends on ALL items), but the sliding-window matheuristic solves local subproblems (window w).

Solution Approach: Multi-Stage Matheuristic:

- **Stage 1:** *Forward-Solve Matheuristic* with or without in-stride load balancing to achieve an initial prioritization-focused solution.
- **Stage 2:** *Post-Processing Matheuristic* to minimally adjust Stage 1 solution to achieve feasibility (both packing feasibility and load balancing).

Key Takeaways

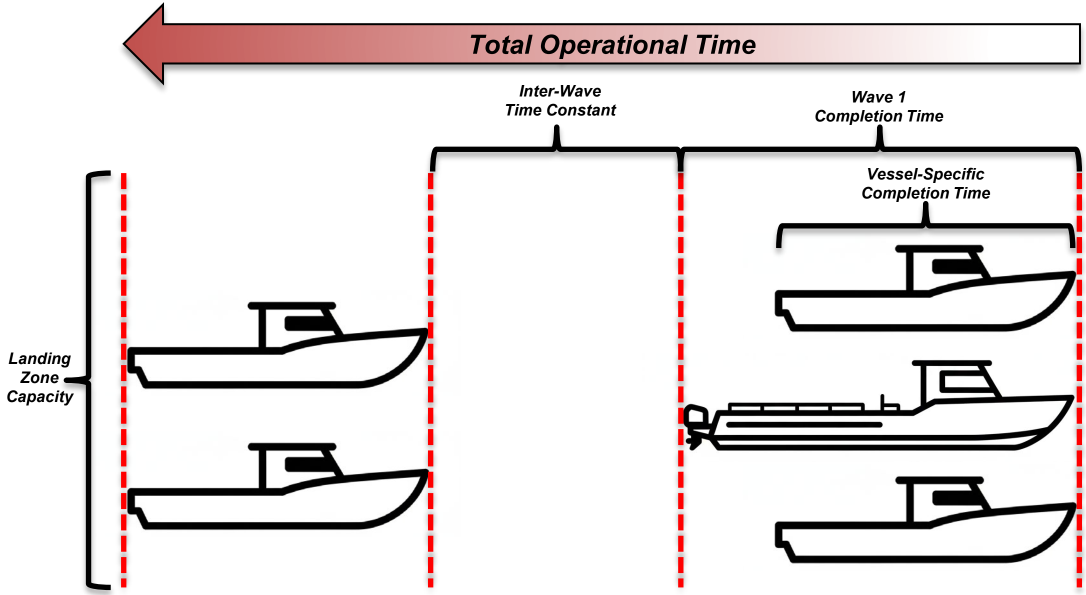
- A **mathematical formulation for combat loading** using prioritized 2D packing that integrates critical operational constraints, such as global load balancing.
- A **robust, multi-stage matheuristic framework** that fundamentally outperforms standard MILP approaches in reliability, solution quality, and computational speed.
- A **method for automatically generating load plans** that are both **operationally feasible** and **optimized for multi-level tactical priorities.**

Paper 3: Multi-Vessel Fleet Optimization

Problem Overview

The Challenge:

- **Considerations:** Which equipment groups → which vessels? Which vessels → which waves? Landing zone capacity constraints.
- **Objective:** Minimize time-weighted priority penalties based on a group's elapsed off-load completion time.



Linear vs. Nonlinear Time:

- **Linear time.** Inter-wave time constant.
- **Nonlinear off-loading time.** A power law model to account for congestion and/or threat level at the landing zone. d^{γ} is the off-loading time for a group relative to its assigned vessel, where:
  - d is the distance of the group to the vessel's off-loading point. and
  - $\gamma \in [1, 3]$ is the congestion factor.
- **Context-Dependent Vessel Selection:**
  - $\gamma \leq 1.5$ (peacetime/permissive environment) → larger vessel for capacity advantage.
  - $\gamma \geq 2.0$ (contested environment) → other vessel types to balance congestion and threat level.

Solution Approaches:

- **Greedy parallel machine scheduling.** Construction heuristic.
- **Metaheuristic improvement.** Group and vessel level operators.

Paper 3: Multi-Vessel Fleet Optimization

Solution Approaches and Key Takeaways

Baseline: Parallel Machine Scheduling Construction Heuristic.

- **Greedy Assignment:** groups → vessels → waves (priority order based on lowest completion time).
- **Strong baseline** for any vessel combination. **Fast:** solves in fractions of a second.

Metaheuristic Refinement:

- **ILS (Iterated Local Search):** *Single-state iterative improvement*.
  - Vessel-level and group-level swapping operators.
  - Fast exploration of local neighborhoods.
- **Memetic Algorithm:** *Population-based search* (n=30 individuals).
  - Solution-level crossover and mutation operators.
  - Lamarckian learning (local improvements inherited from ILS).
- **Both Approaches:**
  - Use *lower bound methods* from previous work as *fast objective proxy*.
  - Avoid computationally expensive full packing/loading solves during search.

Algorithm Comparison Across Operational Scenarios

Scenario	γ (congestion factor)	Memetic Improvement (from baseline)	ILS Improvement (from baseline)	Evals/Hour (Memetic vs. ILS)
Peacetime / Permissive	1.0	***4.72%***	0.73%	187 vs. 75
Standard Operations	1.2	***12.84%***	0.95%	188 vs. 73
Contested Environment	1.5	***16.15%***	1.04%	187 vs. 76
High Threat	2.0	***15.86%***	1.05%	186 vs. 76

Key Takeaways

- **Structural Insights:** Understanding optimal vessel combinations and wave structures across varying vessel types, landing zone capacities, and congestion levels to inform operational planning and portfolio design decisions.
- **Robust Solution Methodology:** Metaheuristic approaches provide a generalizable framework that handles diverse operational levers and enables comprehensive structural analysis through efficient search.

Conclusion

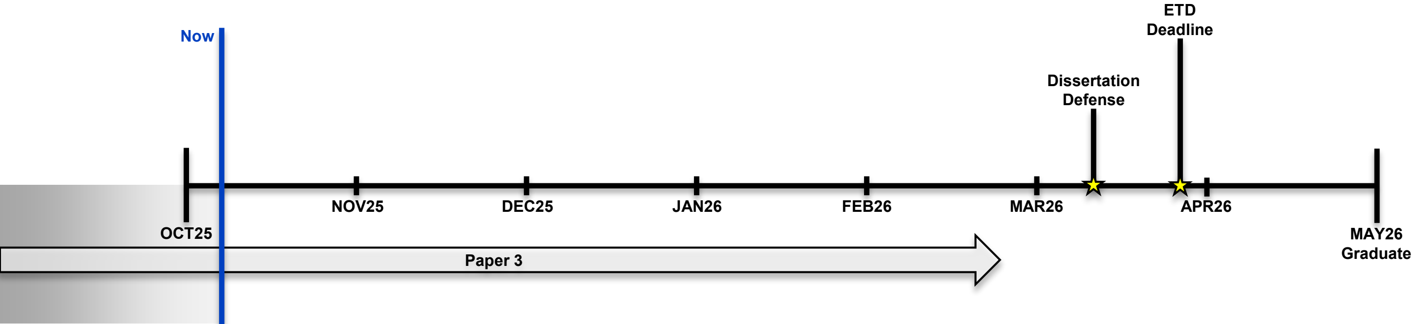
Contributions & Future Work

Key Contributions:

- **Novel prioritized packing framework** integrating spatial layout optimization with multi-level tactical priorities through a flexible prioritization matrix (Papers 1-2).
- **Scalable solution methodologies** combining matheuristic decomposition techniques for single-vessel optimization with population-based metaheuristics for multi-vessel fleet assignment (Papers 1-3).
- **Multi-vessel fleet optimization** with wave-by-wave sequencing, landing zone capacity constraints, and strategic portfolio analysis across vessel types and fleet configurations (Paper 3).
- **Automated combat load planning** transforming labor-intensive manual processes into efficient optimization-driven capability that generates tactically sound, load-balanced solutions in minutes.

Future Directions:

- **Stochastic/robust optimization** to handle operational uncertainties such as dynamic threat assessments and evolving mission priorities.
- **Three-dimensional packing extension** incorporating vertical stacking, height constraints, and multi-deck.



Will Kirschenman
Operations Research PhD Candidate
North Carolina State University